Advanced database management systems

Topic 2: Replication and Log shipping

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Overview

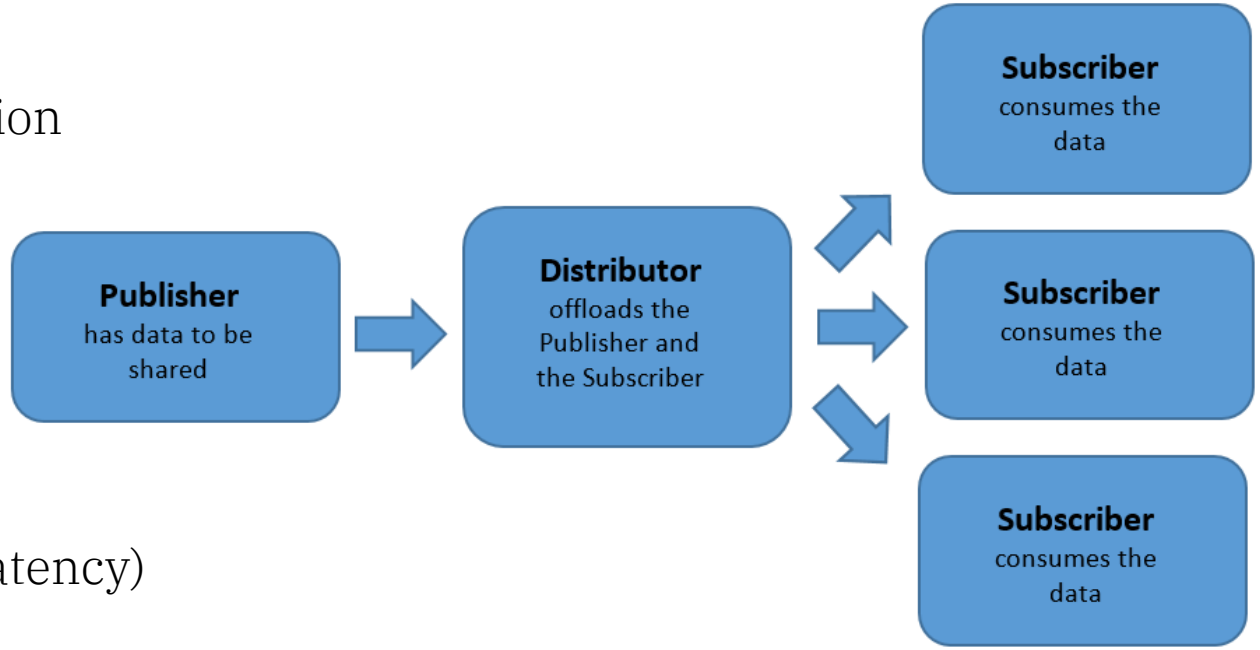
- We automate business workflows with replication solutions
- We design and implement various replication and log shipping configurations

After completing this topic:

- you will understand the basic objectives, use cases and concepts of replication
- you will be able to select the right type of replication for a scenario
- you will be able to design and implement a replication solution with SQL Server
- you will be able to design, start and manage a log shipping solution

Replication conceps

- Replication ~ „copying data”
- The objective is business workflow automation
  - *Not* aimed at improved data security or reliability
  - *Not* a replacement of backups
- Components of the replication metaphor
  - The publisher and the publications
  - The distributor and the subscribers
- Push and pull type subscriptions
- Initialization and regular synchronization (latency)
- The actions performed by agents
  i.e. scheduled jobs



Snapshot Replication

- After initialization, published articles are dropped and re-created at every synch
- Low efficiency due to redundant data transfer
- Not suitable for OLTP subscribers as tables may become inaccessible
- Simple and robust mechanism, based on the BULK COPY technology
- Scenario: data warehousing (ETL) + The first step of any other replication
- Implemented by
  - a shapshot agent writing to the distribution folder
  - a distribution agent reading from the distribution folder

Transactional Replication

- After initialization, the relevant transactions are copied from the transaction log and rolled forward at the subscriber
- Loose coupling (out-of-process) between the publisher and the subcribers
- High efficiency due to only the changed data being transferred
- Ideal and default for OLTP subscribers, low latency
- Scenario: site autonomy, coupling of disparate systems
- Implemented by
  - a shapshot agent for initialization
  - a log reader agent that copies entries in the distribution database of the Distributor
  - a distribution agent reading from the distribution database and rolling transactions forward

Merge Replication

- There is a publisher, but changes committed at any subscriber will be merged across the system -> guaranteed „eventual” consistency
- Records are identified by GUIDs
- Changes are logged by triggers to local log tables, the distribution db contains only history and error info
- Significant resources are needed for communications
- Concurrent changes of the same records may need conflict resolution
- Ideal for distributed applications
- Scenario:travelling agents
- Implemented by
  - a shapshot agent for initialization
  - a merge agent running at each subscriber

Log Shipping

- The goal is to prepare for disaster recovery by running a „warm” backup server
- Based on regular differential transaction log backups and transaction log restores
- Roles: Primary, Secondary, Monitor server
- If the Primary server fails, the warm backup on the Secondary can be started in minutes
- Initialized by a full database backup/restore (recovery mode=FULL)
- Implemented by
  - a backup job running on the Primary (production) server
  - a copy job running on the Secondary server copying the logs to a network share
  - a restore job running on the Secondary server restoring the backup to the secondary database
  - an alert job running on the Monitor server that monitors the other 3 jobs
- Latency is an important consideration